Migration Patterns and their Impact on GDP

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Introduction

For our project, we aimed to gain insight into the patterns of migration of all the people on this Earth. We decided to explore the relationship between GDP and a variety of predictors related to population and migration. We find the connection between statistics and human geography fascinating. One hears about immigration in the news constantly, and we believe that exploring migration with reliably sourced data will provide us with a fresh perspective on such a controversial, polarized topic.

Methods

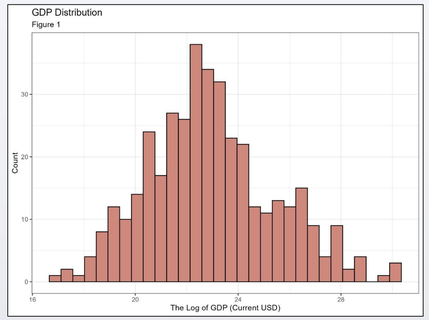
Our migration data came from the Pew Research Center, and our GDP and population data came from the World Bank Group (WBG). Pew is a non-partisan think tank that has long held a reputation as a credible source with excellent statistical methodology. The WBG is the best-known development bank in the world and follows extensive data quality standards developed in conjunction with the International Monetary Fund.

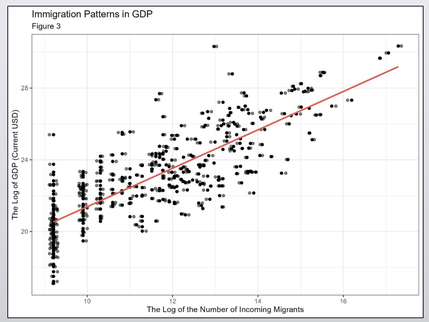
The original migration database contained over 12,000 observations, of which we randomly sampled 8,000 observations over thirty years between 1990 and 2020. Each country appeared multiple times for each religion, so we reduced the data set to only include the rows with the country's majority religion. This reduced the database to 390 observations.

Variable	Units	Min	Median	Max
Log of GDP	Current USD	17.109	22.746	30.342
Log of Population	People	9.215	15.216	21.068
Log of Incoming Migrants	People	9.210	11.290	17.289
Log of Outgoing Migrants	People	9.210	11.918	16.314

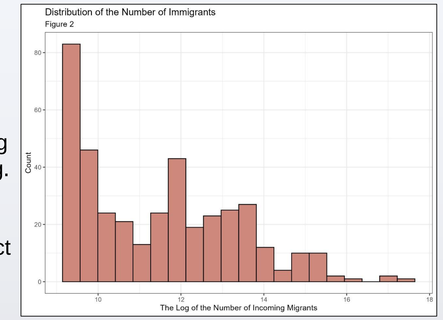
Our response variable was the log of GDP in current USD, a continuous numerical predictor. Our predictors were the year, whether the country has a majority religion or not, the region of the country, the log of the number of incoming migrants, the log of the number of outgoing migrants, the log of the percent of migrants entering the country affiliated with the majority religion, the log of the percent of migrants leaving the country affiliated with the majority religion, and the log of the country's population. We took the log of many of the variables above in order to linearize them. Other than year, which was discrete numeric; majority religion, which was binary categorical; and region, which was nominal categorical, the rest of the variables were all continuous numeric.

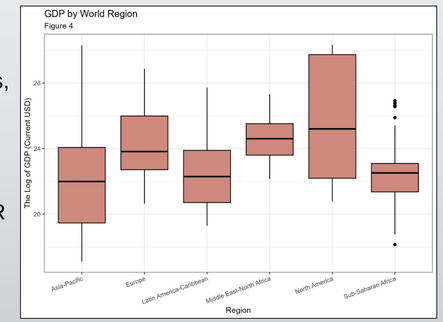
Exploratory Results





The distribution of the log of GDP (fig. 1) appears normally distributed, with a median of 22.746, so the GDP of the median country is $7.6 billion. The distribution of the log of the number of immigrants (fig. 2) is right skewed; most countries had relatively few immigrants compared to a select few countries with many immigrants. There is a strong, positive, linear association between the log of immigration and the log of GDP (fig. 3). As the log of immigration increases, the log of GDP also tends to increase. In figure 4, one can see an association between region and the log of GDP, with high variability in range and IQR by region. North America had the highest median GDP, and Asia Pacific had the lowest median GDP.





Predictive Results

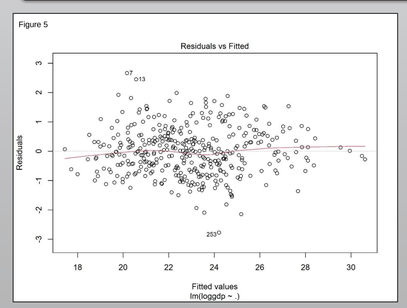
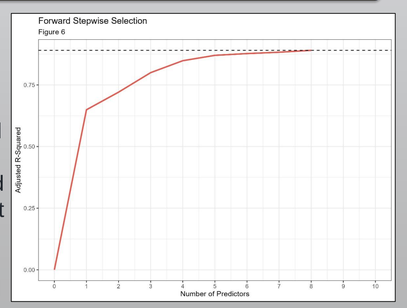
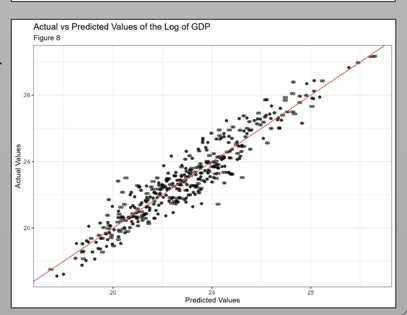


Figure 5 displays the result of taking the log of our variables. The residuals of our linear regression model were evenly distributed above and below zero, indicating that our model followed the linearity assumption. We then used forward stepwise variable selection and found that the best model used all eight variables (fig. 6). Figure 7 displays the 5-fold cross-validated root mean squared error (RMSE) of our linearized model compared to the null and non-linear models. The low RMSE of the linearized model compared to the other models indicated that it was the most accurate model of the three. Figure 8 shows the accuracy of our model by comparing actual GDP values with our predicted GDP values. The values are close to the red line, which represents a perfect model. The tight clustering of the points along the line indicates the accuracy of our model.





Discussion

Our model's performance was far better than that of the null model across multiple error metrics. Based on RMSE, our model was nearly 300% better than the null model. Thus, our model allows us to predict GDP accurately using population and migration data.

Metric	Null Model	Fitted Model
Adjusted R-Squared	0	0.891
Residual Standard Error	2.491	0.824
ANOVA P-value	NA	2.2*10^-16
Cross-Validated Error	2.488	0.833

The biggest technical challenge associated with our project is the collection of data. Finding true population and migration data is infeasible and practically impossible, so they must be estimated. This alone is a complicated statistical problem. There are also barriers associated with gathering migration data, depending on the political climate of a given country at a given time. However, it is also challenging to estimate consumption expenditures, investment, government spending, and net exports – the traditional metrics for calculating GDP. The above problems are all technical, but there are also social difficulties. Any analysis involving migration data could be used unethically or incorrectly to justify policies like immigration restrictions. Bias is also possible depending on how the data is gathered and how illegal immigration is accounted for.

Conclusions

In this project, our exploration and analyses indicate that GDP can be predicted using social factors such as migration and religion. One common critique of GDP is that it doesn't account for non-economic and social factors that influence a country's development and well-being. Our project suggests that there is an association between social factors and a country's GDP. It demonstrates that economic and societal factors of a given country are intertwined and interact with each other.

For future exploration of our topic, one could incorporate more social factors such as literacy rate, disability-adjusted life years, or the World Press Freedom Index as features to the model to see if it improves the accuracy of our GDP predictions. As a result of our analyses, we have gained fresh perspective on the impact of human geography on economics.